

Proposals should involve all actors in the value chain from the manufacturers of net zero technology solutions to energy intensive industries and engineering and construction firms. Interoperability and secure and trusted data sharing between the stakeholders of the value chain should be ensured.

Proposals submitted under this topic should include a business case and exploitation strategy, as outlined in the introduction to this Destination. As project output an elaborated exploitation plan should be developed, including preliminary plans for scalability, commercialisation and deployment (feasibility study, business plan and financial model) indicating possible private and public funding sources (e.g. Innovation Fund, InvestEU and cohesion policy funds). Societal- and environmental impact as well as implications for the workplace (including skills and organisational changes) should be outlined. Proposals should ensure dissemination and replication of the proposed approaches for wide deployment, including advising and building capacity among the relevant actors. Proposals should be based on a sound techno-economic analysis that confirms the economic viability in view of evolving regulatory frameworks.

The inclusion of a GHG avoidance methodology²⁸ is recommended and should provide detailed description of baselines and projected reductions.

The attention of proposers is also drawn to the complementary topic HORIZON-CL4-INDUSTRY-2025-01-TWIN-TRANSITION-05: Advanced manufacturing technologies for leadership of EU manufacturers in products for the net-zero industry.

This topic implements the co-programmed European partnerships Processes4Planet and Made in Europe.

HORIZON-CL4-INDUSTRY-2025-01-TWIN-TRANSITION-37: Solving issues in carbon-neutral iron and steel making processes with diverse input materials of varying quality (Clean Steel Partnership) (RIA)

Call: INDUSTRY	
Specific conditions	
<i>Expected EU contribution per project</i>	The Commission estimates that an EU contribution of around EUR 14.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
<i>Indicative budget</i>	The total indicative budget for the topic is EUR 28.00 million.
<i>Type of Action</i>	Research and Innovation Actions

²⁸ That could follow Innovation Fund methodology: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/innovfund/wp-call/2021/call-annex_c_innovfund-lsc-2021_en.pdf

<i>Technology Readiness Level</i>	Activities are expected to achieve TRL 6 by the end of the project – see General Annex B.
<i>Exceptional page limits to proposals/applications</i>	In order to include a business case and exploitation strategy, as outlined in the introduction to this Destination, the page limit in part B of the General Annexes is exceptionally extended by 3 pages.

Expected Outcome: The topic enables a fast and reliable transition to innovative technology pathways for carbon-neutral iron and steel making by tackling fundamental problems and boundary conditions with a system-level approach. This approach will target input materials, processes, and iron /steel output quality, considering the needs to reduce production costs, find alternative materials and solutions, improve process/energy efficiency and achieve at least the traditional product quality.

Iron and steel making plants constitute complex systems where the product quality is bound to a large set of variables. Variations in feedstock composition, along with reductant choice and mix, introduce noticeable variations in process metallurgy, its kinetics and thermodynamics, with influence on the morphology of the intermediates and consequent impact on the next phases of production, and lifetime, quality, safety and reliability of the finished product.

The strong dependence of the final steel quality on the variable quality of the raw materials and the balance of the production process, has to be taken into account under the consideration of costs, energy availability, sustainability, overall and specific energy efficiency, CO₂ emissions and strategic resources (in particular strategic raw materials²⁹).

Projects are expected to contribute to at least three of the following outcomes:

- Validate innovative carbon-neutral iron and steel making solutions within a system-level approach and in consideration of diverse materials with varying quality (raw input materials and reductants mix) and energy needs. Address high-risk factors at macroscopic and microscopic level through detailed characterisation of the physical and chemical interactions that could compromise the optimal functioning of the processes;
- Solve system-level issues within at least two low-CO₂ production routes;
- Define solutions and provide concepts to address possible modifications or material substitutions in innovative installations for low CO₂ iron and steel production;
- Improve low-CO₂ steel production reliability to target high-quality products: i) clarify the effect of material and process variables, and overall system aspects; ii) clarify the influence of changing crude steel quality on the properties of the produced steel, with the purpose to achieve quality and extended lifespan of products; iii) clarify the impact of diverse input materials with varying quality on the residue characteristics and on its potential valorisation and use;

²⁹

<https://www.consilium.europa.eu/en/infographics/critical-raw-materials/>